Vocabulary resonance in US trademark filings:
A measurement construct for commercial novelty, with validation and applications

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Abstract

The DeDeo prospective/retrospective Kullback–Leibler resonance framework (Barron et al., 2018; Murdock et al., 2017) was developed on cognitive corpora in which the author of each document is the agent whose novelty is being measured: Darwin's reading notebooks, parliamentary speeches of the French Revolution. This paper applies the framework to a qualitatively different corpus type, the goods/services descriptions of US trademark filings, and introduces the resulting construct, denoted ΔKL, as a filing-level measure of commercial novelty that is cheap to compute, available for millions of firms that never patent, and empirically distinct from patent-track invention.

The construct is scored on each filing's topic distribution rather than its raw token counts, following the source papers' own operationalization; this gives every filing a dense representation regardless of length, which matters because goods/services descriptions typically run 10–30 words. All analyses condition on completed registration: completing registration mostly reflects whether the filer followed through to actual commercial use (nine in ten failed applications die by abandonment, not by refusal), so the conditioning separates businesses that launched from filings that remained intentions.

Three results anchor the paper. First, ΔKL marks commercial risk at the mark level, robustly across scoring choices. Among 2016–2018 registrations facing their first Section 8 maintenance gate, marks written in language their industry was moving toward die more often: survival falls -5.5pp across ΔKL quintiles ($n = 770{,}000$; the same to within 0.2pp under token scoring and at four times the topic resolution), and applications register as an inverse-U, with the flux-neutral middle following through to use most reliably. Re-scoring with mixed reference windows shows the gate penalty comes from breaking with the established past, not from resembling the future (-7.3pp against -3.2pp). In either scoring, merely being surprising in the moment is not the signal; the information lives in comparing how surprising a filing is against its past with whether its surprising elements are then taken up by the filings that follow. Second, a methodological caution the construct itself surfaces: under term-level scoring, forward-leaning filings appear to mark better firms (higher margins, more public listings), but these firm-level associations dissolve or reverse under distribution scoring at every resolution tried, and the listing gradient dissolves under document-length controls. Term-level text measures can manufacture firm-performance correlations out of drafting style; distribution scoring is the guard. The firm-level claims are therefore not anchored here, and the within-firm patent contrast (-0.048σ, $t = -7.9$) is read only as evidence that the construct is distinct from patent-track invention. Third, the construct supports direct measurement of cross-industry idea diffusion: software- and tech-services-origin themes arrive in transport and advertising/retail with 1–13 year lags, theme adoption follows

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Bass curves with median $q/p \approx 6.5$, and the cross-class flow is strongly asymmetric (software originates 33 of the 72 origin-to-arrival edges and receives only 10).
The paper closes with a research agenda the construct opens: whether ideas that transfer between industries hold their value, whether suppliers to a new theme outperform its deployers, and how demographic concentration within industries relates to where new themes originate. Code and firm-year panels are public at `https://github.com/ericsilver/novelty`.

1 Introduction

Innovation measurement leans heavily on patents. Patent-based measures are well-validated for technical invention but cover a narrow slice of commercial novelty: most firms never patent, service-sector and business-model innovation rarely patents, and patenting propensity varies so much across industries that cross-industry comparisons are fraught (Dinlersoz et al., 2018; Graham et al., 2013). Trademark filings offer a complementary window. Nearly every commercially serious product or service introduction in the United States is accompanied by a trademark filing, the filing's goods/services description states in plain commercial language what the firm intends to sell, and the corpus extends across every sector of the economy at the rate of hundreds of thousands of granted filings per year. The text of those descriptions, however, has been largely unused; the literature has worked with filing counts and filing metadata rather than filing language.

Murdock et al. (2017) introduced a measurement framework for directional novelty in time-ordered text corpora: for each document, compare its token distribution to the aggregated distribution of past documents (prospective surprise) and future documents (retrospective surprise). The signed difference of those two quantities they call *resonance*: it is positive when a document is unusual against its recent past and close to its near future, negative for the inverse. The framework was developed on Charles Darwin's reading notebooks and extended in Barron et al. (2018) to French Revolution parliamentary debates. Both corpora share a salient property: the text is authored by the agent whose novelty is the object of measurement.

This paper applies the same apparatus to trademark goods/services text, which is commercially incentivised, typically drafted by counsel, and exists to define legal protection scope rather than to express cognition. Whether the resonance framework produces interpretable structure on text of this kind is the threshold question, and the answer this paper documents is yes. The construct, denoted ΔKL, marks commercial risk at the level of the individual mark, robustly across scoring choices. Filings whose vocabulary leans toward where their industry's language is going fare worse as legal objects: their applications complete registration slightly less often (a step that mostly measures follow-through to commercial use, since roughly nine in ten failed applications die by abandonment rather than by refusal), and the registered marks fail the first Section 8 maintenance gate more often. The construct also surfaces a methodological hazard that this paper documents rather than falls into: under term-level scoring, the same forward-leaning filings appear to belong to more successful firms, but those firm-level associations dissolve or reverse under distribution-based scoring at every resolution tried, and the listing gradient dissolves under document-length controls. The signal is uncorrelated-to-negatively correlated with patenting within firm; ΔKL captures a dimension of commercial text that the patent record does not, whatever its ultimate relation to firm performance.

Because the construct is computed from public text with public code, it scales to questions that firm-survey or patent-linked measures cannot easily reach. The paper demonstrates one such application in depth, measuring the diffusion of business vocabulary across industries directly at the phrase and theme levels, and sketches a research agenda: the portability of transferred ideas, the relative fortunes of suppliers versus deployers of a new theme, and the relationship between demographic concentration within an industry and where new themes are born.

Section 2 defines the construct, the corpus, and known fragilities. Section 3 reports the commercial-outcome validation. Section 4 reports the within-firm patent contrast and the expert-list discriminant. Section 5 reports the cross-industry diffusion application. Section 6 reports a base-rate discipline result on entrant-vs-incumbent classification. Section 7 states the research agenda. Section 8 bounds the claims. Section 9 is reproducibility.

2 The construct

2.1 The DeDeo apparatus on trademark text

For a time-ordered sequence of documents, with each document at year t producing a token distribution P_t, the framework defines two quantities against a window W of past and future documents (Barron et al., 2018; Murdock et al., 2017):

- *Prospective surprise*: $\mathrm{KL}(P_t \,\|\, Q_{\text{past}})$, where Q_{past} is the aggregate token distribution of documents in years $t - W$ to $t - 1$.
- *Retrospective surprise*: $\mathrm{KL}(P_t \,\|\, Q_{\text{future}})$, where Q_{future} is the aggregate distribution of documents in years $t + 1$ to $t + W$.

The signed resonance is

$$\Delta\mathrm{KL}_t \;=\; \mathrm{KL}(P_t \,\|\, Q_{\text{past}}) - \mathrm{KL}(P_t \,\|\, Q_{\text{future}}).$$

Positive ΔKL marks a document whose vocabulary is unusual relative to its recent past and close to its near future: the field moved toward this filing. Negative ΔKL marks the inverse, language that looked normal at filing and looked played-out in hindsight: the field moved away.

2.2 Corpus and conditioning

The USPTO Trademark Annual Applications backfile (product TRTYRAP) provides a complete record of US trademark filings from 1884 onward. The deepest diagnostic work in this paper uses the complete software/electronics class (NICE 009; 1.81M filings), and the diffusion application uses four classes selected to exercise canonical cross-industry flow: software (009), scientific and technical services (042), transport (039), and advertising and retail (035), totalling 2.32 million granted filings 1990–2024. The firm-level panels link trademark owners to SEC EDGAR financial filers and to PatentsView patent assignees.

Two conditioning rules apply throughout. The corpus is restricted to *granted* filings (registration date populated), screening out speculative or defensive applications that did not survive examination. And the lifecycle outcomes in Section 3 keep the examination step and the maintenance step separate: registering and maintaining are different commercial acts with different drivers, and pooling them conflates examiner behavior with the use requirement. Mark status is read from the USPTO status-code dictionary directly (700 registered; 701–705 Section 8 accepted; 710 cancelled for Section 8 failure; 800 renewed; 900 expired), and maintenance is measured on registration cohorts for which exactly one Section 8 gate has elapsed. The dictionary-and-cohort design replaces a heuristic outcome coding used in earlier iterations of this analysis chain; the data provide a status snapshot rather than a status history, so the only way to attribute a cancellation to a specific maintenance gate is to restrict to cohorts in which exactly one gate has passed, and the only validated map from snapshot to outcome is the official code dictionary, ground-truthed here against marks with publicly known fates.

2.3 Computation

The construct is scored on topic distributions, following the source papers' own operationalization: the Darwin and French Revolution analyses computed surprise on each document's topic mixture rather than its raw token counts (Barron et al., 2018; Murdock et al., 2017). A latent Dirichlet allocation model with $T = 50$ topics is fit on a stratified sample of 448,437 filings drawn across all 45 classes (vocabulary of 62,168 unigrams and bigrams), every filing 1990–2024 is transformed to a dense 50-topic distribution, and each filing's prospective and retrospective KL are computed against class-year topic references at $W = 5$, scoring 1995–2019. Topic scoring matters here for a reason the source corpora did not face: goods/services descriptions typically run 10–30 words, and on documents that short, token-level KL inflates mechanically (mean $|\Delta\text{KL}|$ varies 2.5-fold across document-size bands), while every filing's topic distribution is dense regardless of length and the artifact largely disappears. A $T = 200$ refit confirms the headline results are not sensitive to topic resolution.

A term-level variant is retained alongside: token distributions (unigram + bigram, within-class document frequency ≥ 50) against the same class-year windows, with Dirichlet smoothing ($\alpha = 1$ per cell), computed at $W \in \{3, 5, 7\}$. The term variant is used where vocabulary identity is the object (the phrase-transit analysis, the reference-window decomposition, the months-scale freshness measure), and as a comparison scoring throughout; because its two windows need not match, it also composes into asymmetric constructs (prospective surprise against one window, retrospective against another) that Section 3.2 uses to ask *which* novelty the outcomes respond to.

The signal depends on having enough past and future filings to populate the references. Figure 1 shows the reference-window density by year for two diagnostic classes; the 5-year past window is incomplete before 1990 and the 5-year future window is incomplete after 2020, and both zones are excluded from the construct's interpretable range. Within the supported window the classes analyzed here have between ∼3,500 and ∼80,000 filings per year.

The burn-in cut is set empirically rather than by the window arithmetic alone. Thin past references inflate prospective surprise, so contamination appears as a positive bias in the per-year mean of signed ΔKL relative to the class's long-run level. Profiling that bias across all 44 classes with sufficient volume: the median absolute bias is ∼2.0 nats in 1990, 0.25 in 1991, 0.10 in 1992, and 0.04 from 1993 onward, below any effect size this paper interprets. The deviations that reappear in 1995–1997 (∼0.15–0.17) are not burn-in, because mechanical contamination cannot rebound; they are the dot-com era moving the corpus, which a longer burn-in would discard as data. Per-class burn-in lengths were also estimated under a stability-band criterion; the apparent cross-class differences track each class's noise floor rather than its reference density (correlation of burn-in length with early filing volume +0.26, with a slow-moving-class proxy +0.14), so there is no empirical case for industry-specific burn-ins. All analyses therefore use a single standard cut at 1995, the first scored year whose full 5-year past window lies inside the dense corpus, which the bias profile shows is conservative by about two years.

2.4 The construct at the case level

Figure 2 grounds the construct in recognizable examples. The two scatter axes are prospective surprise and retrospective surprise, with ΔKL as the diagonal residual. Filings below the diagonal are *innovator*: the language looked unusual at filing and became standard afterwards. Filings above the diagonal are *tired*: the past resembles the filing more than the future does.

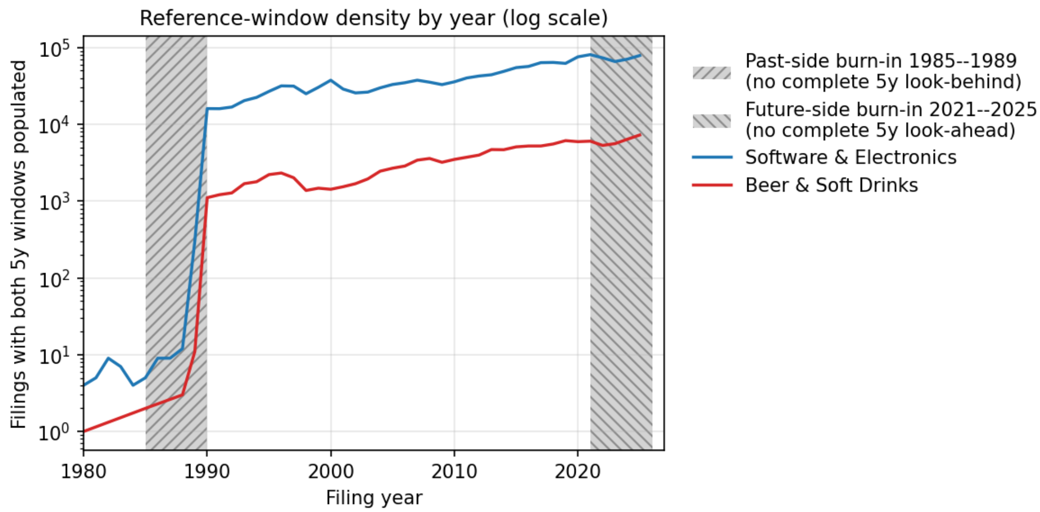

Figure 1: Reference-window density by year on log scale. Hatched regions are the past-side and future-side burn-in zones where the 5-year window is incomplete.

2.5 Known fragilities

The construct has documented fragilities on this kind of text, and they motivate the choice of topic scoring as the method. Prospective and retrospective KL are highly correlated however scored (term scoring: $r = +0.971$ on the complete Class 009 corpus; topic scoring: 0.950), so signed ΔKL is a small residual of two large, stable quantities; what has not been seen before will mostly not be seen again, and the signal lives in the difference. Under term scoring the residual is noisy at the per-filing level, and its magnitude inflates mechanically for short filings (mean $|\Delta\text{KL}|$ of 0.293 for documents under 5 terms, decaying to 0.147 at 40 or more, a 2.5-fold gradient); under topic scoring the gradient falls to $\sim$1.4-fold, which is why topic scoring carries the outcome analyses. The two scorings agree only moderately per filing (Pearson 0.46, Spearman 0.36 on 1.9M jointly scored filing-class observations), which is what gives their agreement on the mark-level outcome patterns evidential force, and their disagreement on the firm-level patterns diagnostic force (Section 3.3). Pre-1995 readings are contaminated by thin past references. ΔKL is robust to the duration of the timing window considered: across $W \in \{3, 5, 7\}$ years, only 6.9–10.4% of filings flip sign, well under conventional fragility thresholds. The per-filing noise attenuates at every aggregation used in this paper: firm-year means, token-level pools, and theme-level prevalence trajectories.

3 Validation against commercial outcomes

If ΔKL measures something commercially real, it should predict outcomes that cost firms money. At the *mark* level it does, robustly: the trademark lifecycle selects against forward-leaning vocabulary, whose applications complete registration slightly less often and whose registered marks fail the first maintenance gate more often, under topic scoring at two resolutions and under term scoring alike. At the *firm* level the corpus teaches a different lesson. Term-level scoring suggests that firms filing forward-leaning marks do better (listing, margins); distribution-based scoring erases or reverses those associations at every resolution tried. Section 3.3 documents the divergence and reads it as a methodological hazard for text-based novelty measures generally, rather than asserting either sign as a property of firms.

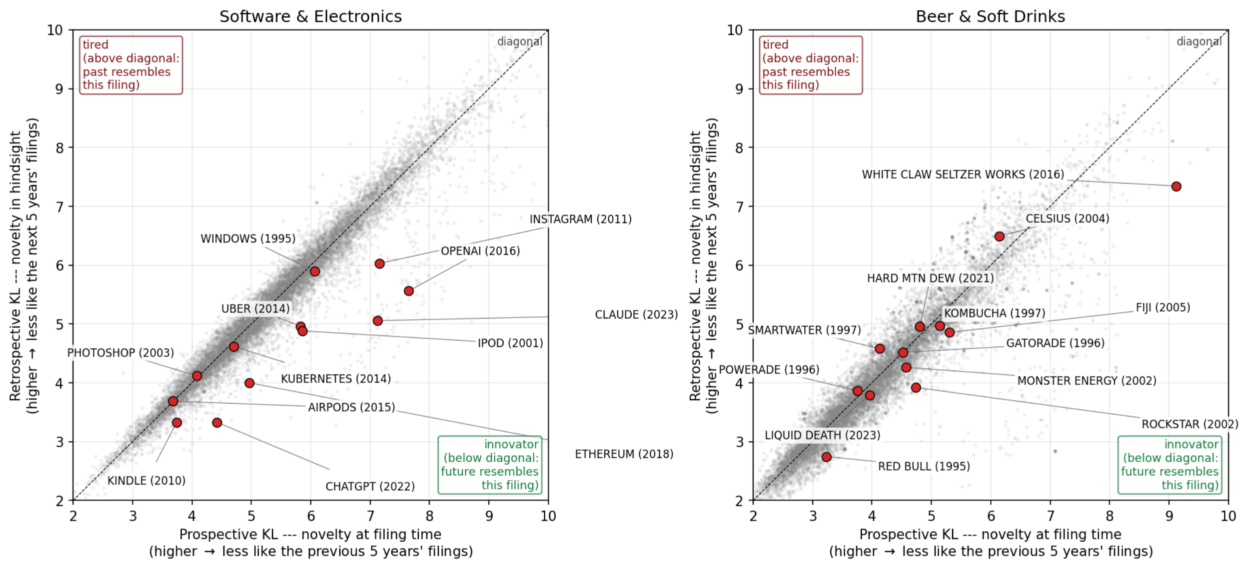

Figure 2: Prospective KL versus retrospective KL with ΔKL as the above-diagonal/below-diagonal residual. Labeled examples are recognizable post-burn-in brand introductions, positioned by their goods/services text. *Left:* software/electronics. OpenAI (+2.09), Claude (+2.06), Instagram, ChatGPT, iPod, Ethereum, and Uber sit on the innovator side; Photoshop (2003) and AirPods sit essentially on the diagonal. *Right:* a beer/soft-drinks diagnostic class. White Claw (+1.78), Rockstar, and Red Bull lean innovator; Smartwater (1997) and Celsius (2004) lean tired.

3.1 Mark-level selection: registration and the first maintenance gate

Outcome definitions. Outcomes derive from the USPTO status-code dictionary. A filing reached registration if its registration date is populated. A registered mark faces the Section 8 use-declaration gate in years five to six; *Cancelled–Section 8* (status 710) marks failure, and *Section 8 accepted* (701/702, partial variants 703–705) or subsequent renewal (800) mark survival. Because older registrations face repeated gates, the gate analysis uses the 2016–2018 registration cohort, for which exactly one Section 8 gate has elapsed by the 2025 data snapshot.

Registration completion is better read as a launch indicator than as an examiner verdict: among applications that fail to register, roughly nine in ten die by abandonment (44% never file the statement of use that registration requires; 45% stop responding during prosecution) and only a few percent lose adversarial or appeal proceedings. Completion (Figure 3A; 3.59 million debut filings, 57.6% base rate) is an inverse-U in ΔKL: the middle quintiles complete best ($\sim$58.6–58.9%) and both tails complete less (54.5% at the bottom quintile, 56.8% at the top), with the same hump in the topic-KL levels. On the launch-indicator reading, the inverse-U says that applicants filing category-typical language most reliably follow through to commercial use, while both kinds of vocabulary bet fail to launch more often. The shape is scoring-robust (term scoring gives depth +4.8pp on all filings pooled; topic scoring +3.7pp, and +3.0pp at $T = 200$) and is the rule rather than a pooled artifact: estimated class by class under term scoring, the middle-versus-tails depth is positive in 37 of 44 classes with sufficient volume and statistically distinguishable from zero in 33 (per-class figure and table in the repository).

The registration inverse-U is also why analyses that skip the registration conditioning see an inverse-U in downstream outcomes. Any unconditional success measure, mark survival included, is the product of the registration rate and the conditional rate, and the registration factor stamps its

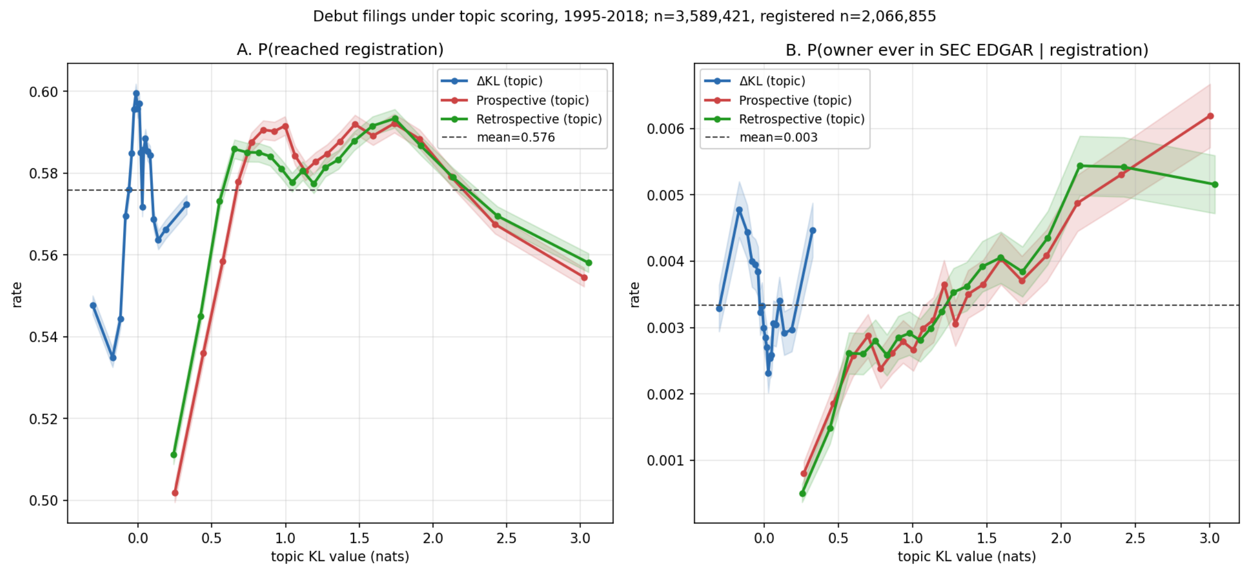

Figure 3: Outcomes for debut filings (owner's first-ever filing) across all 45 NICE classes under topic scoring, filing years 1995–2018, in quantile bins with 95% intervals. $n = 3{,}589{,}421$; registered subset $n = 2{,}066{,}855$. *A:* P(reached registration). *B:* P(owner ever in SEC EDGAR | registration); the absence of a ΔKL gradient in panel B is discussed in Section 3.3.

middle-favoring shape onto the composite: the flux-neutral middle registers best, so it appears to "survive" best unconditionally even where the conditional gate pattern says otherwise. Appendix A reproduces the outcome analysis without the registration conditioning to make the composite shape visible.

The first Section 8 gate is where forward-leaning marks pay. Figure 4 shows the pooled result on 769,704 gated registrations: survival falls −5.5pp across topic-ΔKL quintiles (51.7% at the bottom to 44.5% at the fourth, with a small recovery to 46.2% at the top; SE 0.2pp). The result is indifferent to scoring: −5.7pp under term scoring on the identical sample, −5.5pp at $T = 200$. Under term scoring the gate survival *rises* in the raw KL level (+6.5pp prospective, +7.6pp retrospective): distinctive text survives the use requirement better; forward-leaning text survives it worse. A natural mechanism runs through the use declaration itself: Section 8 requires continued use in commerce, and vocabulary that bets on where the category is going belongs disproportionately to products that had not yet materialized at filing and sometimes never do.

The gate result is broad but not universal. Per class (Figure 6, term-scored; the topic-scored version is in the repository and agrees), the ΔKL top-versus-bottom quintile lift is negative in 31 of 41 classes with sufficient gated volume under term scoring and 34 of 41 under topic scoring, with the largest negative lifts in physical-goods classes (medical apparatus −21.7pp, tobacco −24.1pp, lighting −19.9pp, machinery −18.4pp) and small positive lifts concentrated in services (legal services +7.4pp, telecommunications +6.3pp). One reading is that a forward-leaning physical-goods filing requires a product to exist by the gate year while a forward-leaning services filing can be serviced with capacity on hand; testing the gradient against class-level production lead times is a follow-up computation.

Summarizing the cross-industry picture: the registration inverse-U is near-universal (37 of 44 classes), while the gate penalty is the majority pattern with a coherent minority. The gate exceptions are not noise scattered across the forest; they cluster in service classes where the cost of

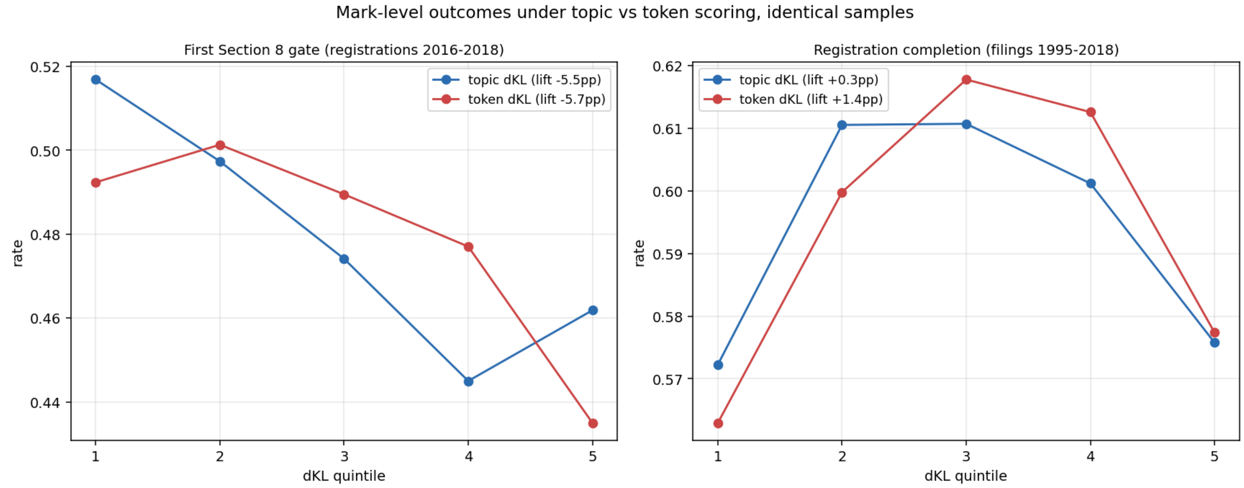

Figure 4: The two mark-level outcomes under topic and term scoring on identical samples, pooled across classes. *Left:* first Section 8 gate survival (registrations 2016–2018). *Right:* registration completion (filings 1995–2018). Both patterns are scoring-robust.

making a forward-leaning filing true by the gate year is lowest. One small class, lace and embroidery (026), shows a genuine mirror image, with tails registering better and forward-leaning marks passing the gate more often; every other class fits the headline pattern on at least one of the two steps.

3.2 Which novelty? Decomposing the gate penalty by reference window

"Forward-leaning" is relative to a reference, and the choice of reference window selects what kind of novelty is being measured. Scoring the full corpus at $W \in \{3, 5, 7\}$ and mixing windows across the two sides gives five variants: three symmetric, plus a foresight configuration (prospective against the recent 3-year past, retrospective against the distant 7-year future) and a past-rupture configuration (prospective against the established 7-year past, retrospective against the near 3-year future).

The gate penalty is monotone in the mix (Figure 5). Pooled across classes on the 2016–2018 gate cohort, the top-versus-bottom quintile lift runs -3.2pp for the foresight configuration, -4.7pp at symmetric $W = 3$, -5.7pp at the $W = 5$ baseline, -6.1pp at $W = 7$, and -7.3pp for past-rupture, which is also the most pervasive variant across industries (negative in 35 of 41 classes, against 31 for the baseline). In software the decomposition is complete: the foresight configuration carries no gate penalty at all (-0.3pp) while past-rupture carries -12.7pp. The Section 8 risk attaches to breaking with the established past, not to resembling the future; lengthening the past window concentrates the penalty and lengthening the future window dilutes it. This is consistent with vocabulary leaving the corpus gradually rather than abruptly: the long-past reference holds the fading establishment, and distance from that establishment is what the use requirement punishes. The registration inverse-U, by contrast, is indifferent to the window mix (middle-versus-tails depth $+4.6$ to $+4.9$pp across all five variants), so the examiner and the marketplace are responding to different aspects of the same text.

The decomposition extends below the year. Scoring class 009 at monthly resolution against two adjacent references, the twelve months ending one month before filing and the twelve months before those, the two surprise levels are nearly identical per filing ($r = 0.992$, the stability property again), but their difference, a months-scale freshness measure that is high when a filing resembles the very recent months far more than the year before them, reproduces the risk pattern: marks riding a wave that emerged within the prior year fail the gate at -9.4pp across freshness quintiles, a stronger separation than the symmetric ΔKL baseline delivers in the same class. Two things follow. Month-scale position in a wave matters, with the penalty concentrated on the months-late rather than the months-early. And because freshness is built entirely from pre-filing text, it is computable on the filing date with no look-ahead, which makes it the construct family's real-time member. This is a single-class result and its corpus-wide estimation is owed.

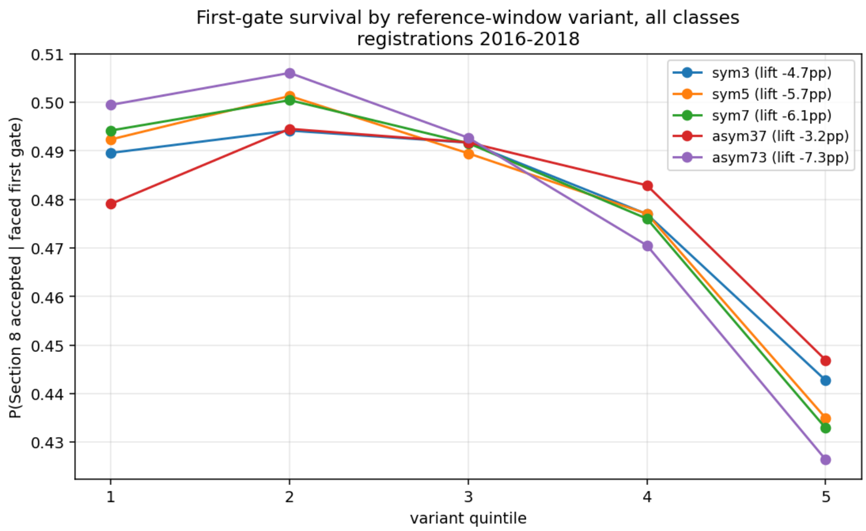

Figure 5: First-gate survival by quintile of each reference-window variant, pooled across classes, registrations 2016–2018. The penalty deepens monotonically from the foresight configuration (pros W=3, retr W=7) to the past-rupture configuration (pros W=7, retr W=3).

3.3 Firm-level associations are scoring-dependent

Under term-level scoring, forward-leaning filings appear to belong to better firms. Among registered debut filings, the probability that the owner ever appears in the SEC EDGAR listed universe rises in term-ΔKL from 0.29% at the bottom quintile to 0.42% at the top; on an SEC-matched panel of firm-years, trailing 3-year mean term-ΔKL predicts gross margin at +2.3pp per σ ($t = 2.7$ on the locally rebuilt panel through 2019; +3.2pp, $t = 5.3$, on the original panel through 2024), with prospective surprise entering positively and retrospective negatively, the resonance configuration; and firm-year term-ΔKL predicts excess stock returns of +1.5 to +5.0pp per σ at one-to-four-year horizons, decaying by year five, with prospective surprise alone predicting approximately zero at every horizon. A larger debut-only return figure (+28pp at 4 years) does not survive a sample-size diagnostic and is not claimed.

None of these firm-level associations survives distribution-based scoring, and the paper therefore does not anchor on them. Under topic scoring on identical samples, the debut listing gradient is flat to declining (Figure 3B), and the margin coefficient reverses: -1.3pp per σ ($t = -1.8$) at $T = 50$ and -2.2pp ($t = -3.1$) at $T = 200$, with the prospective/retrospective decomposition flipping sign. The resolution test is the informative part: if topic scoring had merely coarse-grained away a real term-level signal, quadrupling the topic count should move the estimates toward the term-level result. It moves them further away. Meanwhile the term-level listing gradient dissolves under document-length controls: within document-size bands it is U-shaped rather than rising, with the apparent monotone rise contributed by composition across lengths (listed firms' counsel write longer descriptions; the only large within-band effect is in 40-plus-term filings). The parsimonious reading is that the term-level firm associations ride on drafting-style correlates of firm type, length chief among them, exactly the channel topic scoring neutralizes. The mark-level results of Section 3.1 are unaffected: they hold under both scorings at both resolutions.

This is a finding about measurement with reach beyond this corpus: term-resolved text-novelty measures can manufacture firm-performance correlations from drafting style, and a distribution-based rescoring is a cheap, decisive diagnostic. What does *not* explain the firm-level divergence is the registration filter. Among applications that fail to register, only 4.3% are removed adversarially (appeal, opposition, or inter partes proceedings); 42.6% never file the statement of use and 45.3% stop responding during prosecution, applicant decisions rather than examiner ones. Dropping the registration conditioning entirely leaves the term-level listing gradient essentially unchanged (0.26% to 0.38% across quintiles against 0.29% to 0.42% conditional), so neither the term-level association nor its dissolution is an examiner artifact; and an examiner channel that touches under two percent of applications cannot generate the several-point quintile gaps of the registration inverse-U either.

If a genuine firm-level effect exists beneath the artifacts, the panels here cannot establish it, and two causal readings would still compete: a realized-business channel (the trademarked line itself pays) and a propensity channel (firms disposed to bet early both file forward-leaning marks and perform). The discriminating evidence sits in applications that never registered, which carry the propensity signal without the realized line; that test, and length-controlled re-estimation of the margin and returns panels, are the firm-level follow-up work this section motivates.

4 Distinctness from patents

If ΔKL and patent counts were noisy measures of the same underlying construct, they should correlate within firms over time. They do not.

The within-firm analysis uses a published panel of 174,569 firm-years of trademark owners name-matched to PatentsView patent assignees (PatentsView, 2024); the fixed-effects estimation uses the 94,106 firm-years across 14,470 firms with within-firm variation. The panel and matching code are in the public repository. The matching normalizes both sides (strip corporate-form suffixes and punctuation, lowercase) and joins on exact normalized-name equality. One provenance note: the published panel was constructed under an earlier exponential-decay kernel variant of the construct; its firm-year mean ΔKL correlates at $r = 0.87$ with the current flat-window scoring on 68,013 matched owner-years, so the results below are expected to be kernel-stable, and re-estimation on the current scoring is owed with the next panel release. I fit a two-way (firm + year) fixed-effects regression of standardized firm-year mean ΔKL on log(1 + patents) with firm-clustered standard errors.

The coefficient is negative and precisely estimated, strengthening under filing-count control. The strongest negative coefficient sits on patents_{t-1}: in years after a firm patents heavily, its trademark vocabulary regresses toward class-typical. A reading consistent with the data is temporal substitution between patent prosecution and market-facing brand work; either way, the two measures are not interchangeable.

The contrast with expert judgment completes the discriminant picture. Across the 7,431-firm public panel, firms appearing on the BCG Most Innovative lists (2021–23) sit $+0.28\sigma$ above the panel mean in ΔKL ($n = 31$ matched, $p = 0.012$), firms on the MIT Technology Review 50 Smartest Companies list (2015) sit $+0.50\sigma$ above ($n = 17$, $p = 0.0006$), and the pooled union sits $+0.34\sigma$ above ($n = 43$, $p = 0.0003$). Patent counts are uncorrelated with ΔKL on the same panel (Pearson

Table 1: Within-firm ΔKL versus patenting. Coefficients in standard deviations of ΔKL per unit log(1 + patents).

Specification	coef (σ)	SE	t
Between-firm correlation (firm means)	−0.020	—	—
Within-firm correlation (firm+year demeaned)	−0.015	—	—
Two-way FE, contemporaneous	−0.048	0.006	−7.9
robustness: patents = max over matched assignees	−0.048	0.006	−8.0
controlling log(n_filings)	−0.051	0.006	−8.4
Lead/lag: patents_{t-1}	−0.054	0.009	−5.8
patents_{t}	−0.019	0.010	−1.9
patents_{t+1}	+0.010	0.010	+1.0

+0.010). The expert lists reward something patents do not capture, and term-scored ΔKL tracks it. The matched samples are small, the panel is term-scored and so inherits the firm-level scoring caveat of Section 3.3, and the result is a discriminant check, not a headline.

5 Application: cross-industry diffusion, measured directly

Because ΔKL is computed from open text, the same corpus supports measuring where business vocabulary is born and how it travels. This section demonstrates the construct's reach beyond firm-level validation.

5.1 Phrase-level transit

For 14 curated business-model phrases, I record the first year each of the four classes crosses a 1% prevalence floor among granted filings.

Table 2: First year a theme reaches 1% prevalence among granted filings, by class. "–" = never clears the floor.

Theme	software	tech-svcs	transport	advert/retail
as-a-service	2011	**2009**	2013	2012
mobile-app	2010	**2009**	2012	2012
cloud	2012	**2010**	2011	2014
streaming	**2008**	2008	2013	2014
artificial-intelligence	2017	**2016**	2018	2018
augmented/virtual-reality	**2015**	2016	2022	2022
blockchain / crypto	2021	**2018**	–	2022
real-time	**2001**	2003	2009	2014
on-demand	2017	2016	**2014**	2017

Software- and tech-services-origin phrases arrive in the physical-economy classes with lags of one to thirteen years; the only counterexample in the set is on-demand, which originates in transport. The 1% floor is sensitive to class size (transport has roughly one-eighth the volume of software), and the asymmetry is a four-class result that a 45-class build would test properly; the current evidence is consistent with the directional reading but cannot exclude a corpus-construction alternative.

5.2 Theme-level structure

A theme-level analysis with unsupervised topic extraction corroborates the phrase-level pattern without depending on which phrases were picked. I fit LDA with $T = 50$ on a stratified 200,000-filing sample (vocabulary $V = 69{,}361$) and transform across the full 2.32M-filing pool. The resulting topics are recognizable business-model components (health/wellness; autonomous loading and material handling; retail store services; electronic transmission and computer networks; vehicles and traffic).

For each (theme, class) cell that reaches the 1% floor and is tracked at least six years, I fit the Bass diffusion model (Bass, 1969). Of 118 retained fits ($R^2 \geq 0.7$), the median innovation coefficient p is 0.018, the median imitation coefficient q is 0.110, and median $q/p \approx 6.5$, in the range typically reported for consumer-product adoption. Most fits are platform-like; a small tail is fad-shaped (renewable-energy and virtual-reality themes show $q/p > 40$, with steep adoption and visible recession).

Tagging each multi-class theme's origin class (earliest to cross the floor) and counting one directed edge from the origin to each later arrival class gives 72 origin-to-arrival edges over the 33 multi-class themes. Software (009) is the dominant origin, contributing 33 of the 72 edges against the 18 expected under symmetry, and receiving only 10; the software-plus-tech-services pair sends 26 edges to transport and advertising/retail and receives 18 back. The same asymmetry reproduces under independently-fitted NMF on the same sample and vocabulary, with seven of ten curated seed phrases recovering a clear best topic. Both methods are bag-of-words factorizations, so this robustness is within-paradigm; a sentence-embedding or graph-community extraction is the natural cross-paradigm test and is not run here.

6 Do new themes come from incumbents or entrants?

The corpus invites population-share questions, and it punishes them when the base rate is ignored. As a worked example with substantive interest of its own: when a new theme appears, is it carried by new entrants or by incumbents diversifying in? Schumpeter (1942) introduced the distinction (Mark I entrant-led, Mark II incumbent-led), and the standard empirical form reports the entrant share among early adopters.

Applying the standard form to the 50 LDA themes, the 50 earliest carriers per theme are 73.7% entrants on average, and 48 of 50 themes are majority-entrant. The raw numbers look like strong entrant-led adoption. But the corpus's debut rate among all granted filings in this period is 76.4% under year-matched weighting; most filings are by owners debuting that year. Against that baseline the theme-carrier excess is −2.7pp, and 28 of 50 themes sit below baseline. Themes inherit the corpus's high debut rate without adding to it.

The residual pattern is the interesting part. The themes carried disproportionately by incumbents relative to baseline include environmental/sustainability, natural gas and petroleum, artificial intelligence, and cloud computing. The canonically new technology themes of the 2010s and 2020s lean incumbent: established firms diversifying into AI and cloud pull the vocabulary signal forward at least as much as de-novo entrants do. This is consistent with the within-firm patent contrast of Section 4; the firms moving the vocabulary are not necessarily the firms moving the patents.

AI is not (yet) an entrepreneurial explosion

The corpus supports a direct comparison of the AI era to the one episode of genuine vocabulary rupture in the record. Aligning the two eras at their starts (internet 1994, AI 2015) and tracking era-term prevalence in the software and tech-services classes: through year four the two track each other almost exactly (internet 7.5% of filings, AI 4.6%), and then they diverge. Internet vocabulary tripled between years four and six, reaching 23% of filings in 2000; AI vocabulary has compounded steadily to 12.9% at year nine, roughly where the internet stood at year five (Figure 7). The release of consumer chatbots is visible as an acceleration (6.8% in 2022 to 10.2% in 2023), not a discontinuity. Corpus turbulence tells the same story: the per-year mean of prospective surprise, which spiked along with internet vocabulary in the late 1990s relative to its trend, shows no AI-era deviation at all through 2024. Combined with the incumbent lean of the AI theme above, the trademark record's current verdict is that AI is an incumbent-led vocabulary adoption, not an entrepreneurial explosion. The falsifiable forward claim: if AI follows the internet's path with a lag, its prevalence should triple within roughly three years of this paper's data edge, and the entrant share of its carriers should rise above the corpus baseline rather than sit below it.

7 A research agenda the construct opens

Each of the following is computable from the public corpus and panels with the apparatus of this paper. None is run to completion here.

Are transferred ideas as valuable away from home? The flow graph identifies, for every theme, its origin class and arrival classes. Linking carrier filings to the SEC panel tests whether a theme earns the same margin and survival premia in destination industries as in its origin. A first-pass within-firm estimate finds a contemporaneous gross-margin lift of +1.4pp per σ of borrowed novelty ($t = 2.1$) that fades within a year; whether portability is the rule or the exception is open.

Suppliers versus deployers. When a phrase crosses industries, the carrier filings split into firms whose offer *is* the new thing and firms whose offer *supplies* the firms doing the new thing, the sellers of pick-axes rather than the panners for gold. The goods/services text is rich enough to separate the two roles, and the gold-rush intuition that suppliers outperform deployers is testable against renewal, margins, and listing.

Demographic concentration and where themes are born. Surname-imputed demographic composition of filing owners, applied to US-domiciled individual filers, identifies industry enclaves in which a demographic group is heavily over-represented. Preliminary work finds enclave filings sit on the innovation tail of ΔKL rather than the harvest tail, and that owners who diversify across classes list publicly at many times the single-class rate. Whether enclave industries disproportionately originate themes, and whether enclave-rooted firms carry them outward, is a direct join of the demographic panel to the flow graph.

External codification. The USPTO ID Manual periodically standardizes acceptable goods/services wording. If high-ΔKL filings use terms the Manual later codifies, while those terms are still rare in class, the construct anticipates institutional recognition, the strongest available external validation and one that requires only the Manual's version history.

Cross-paradigm theme extraction. The diffusion structure has been shown method-stable within bag-of-words factorizations. A sentence-embedding clustering or graph-community extraction on the same corpus would establish (or refute) paradigm independence.

8 Scope and limits

The paper makes four substantive claims. ΔKL, computed as in Barron et al. (2018); Murdock et al. (2017) on topic distributions of USPTO trademark filings, produces interpretable structure on commercially-incentivised legal text. It marks commercial risk at the mark level, robustly across scoring choices: forward-leaning applications complete registration slightly less often (an inverse-U with the flux-neutral middle following through best) and forward-leaning registered marks fail the first Section 8 maintenance gate more often (-5.5pp across quintiles pooled, the same under term scoring and at $T = 200$; negative in 31–34 of 41 industries depending on scoring). It is empirically distinct from patent-track invention within firm. And it supports direct measurement of cross-industry vocabulary diffusion, with multi-year origin-to-arrival lags, Bass-fittable adoption, and strongly asymmetric flow out of software.

The paper rejects three readings the evidence does not support. ΔKL does not measure innovation in any strong sense; the defended primitive is lexical resonance on commercial text. ΔKL does not identify an exploitable extreme-tail return premium; the large debut-specification figure fails a sample-size diagnostic. And the term-scored firm-level advantages of forward-leaning filings (margins, listing) are not asserted as properties of firms: they dissolve or reverse under distribution scoring at every resolution tried and under document-length controls, and the paper reads them as drafting-style artifacts pending evidence to the contrary.

The Section 8 gate result is identified on a single registration cohort (2016–2018, the cohort for which exactly one gate has elapsed), so cohort-specific shocks, including the pandemic years falling inside the gate window, are not separable from the ΔKL gradient; replication on the 2019–2021 cohort as its gate completes is the direct check. The theme-level diffusion analysis is a four-class build; per-class ΔKL scores now exist for all 45 classes, but the theme panel and flow graph have not been re-estimated on the full set. The expert-list discriminant (term-scored, $+0.34\sigma$, 43 matched firms) inherits the firm-level scoring caveat and is read as suggestive only.

Questions owed to follow-up work: the four agenda items of Section 7; whether the mark-level gate penalty reflects vocabulary position itself or selection on owner covariates; whether the cross-industry gradient in the gate penalty tracks production lead times or vocabulary churn; length-controlled re-estimation of the firm-level panels and the never-registered-applications test of Section 3.3; and corpus-wide estimation of the months-scale freshness measure.

9 Reproducibility

Code, firm-year panels, and analysis outputs are available at `https://github.com/ericsilver/novelty`. The pipeline rebuilds from a USPTO Open Data Portal API key with a single `make tm` command (the underlying TRTYRAP backfile is approximately 12 GB streamed once). The published panels include the SEC-matched firm-year ΔKL panel ($n = 59{,}033$) and the PatentsView-joined panel ($n = 174{,}569$). Analysis scripts and their outputs are indexed in the repository `README.md`.

A The outcome analysis without registration conditioning

Figure 8 repeats the lifecycle outcome analysis without the registration conditioning, on filings 2013–2015 pooled across all classes (a window whose registrations land by $\sim$2017 and face at least one Section 8 gate by the 2025 snapshot; the earliest registrations in the window also face their year-ten renewal, so panel B mixes one and two gates). Panel A is the unconditional composite, the probability that a filing both registers and passes the gate, measured over all filings including the never-registered. It is an inverse-U in ΔKL (quintiles 22.0%, 22.9%, 23.0%, 22.5%, 21.4%; middle-versus-tails depth +1.4pp, SE 0.1pp), because the composite inherits the registration step's middle-favoring shape from Figure 3A. Unconditional survival analyses on this corpus therefore recover an inverse-U whether or not the maintenance gate itself favors the middle, which is why the body of the paper conditions on registration throughout. Panel B shows the conditional gate outcome on the same filings for contrast; on this mixed one-to-two-gate cohort it is mildly inverse-U with a clearly penalized forward tail, consistent in the right tail with the cleaner single-gate cohort of Figure 4 and a reminder that the left tail of the gate result is cohort-sensitive.

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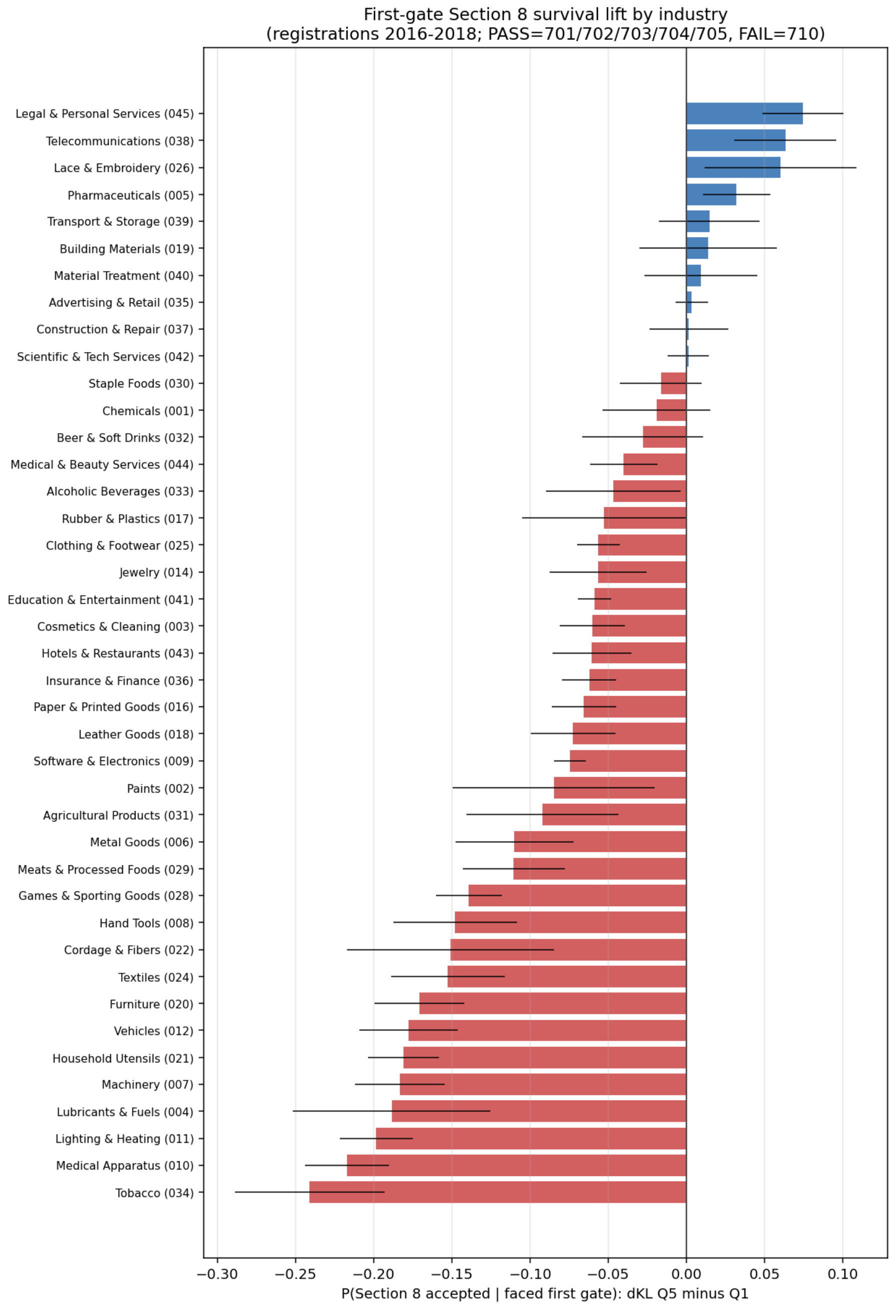

Figure 6: First-gate Section 8 survival lift (ΔKL top minus bottom quintile) by industry, registrations 2016–2018, with 95% intervals.

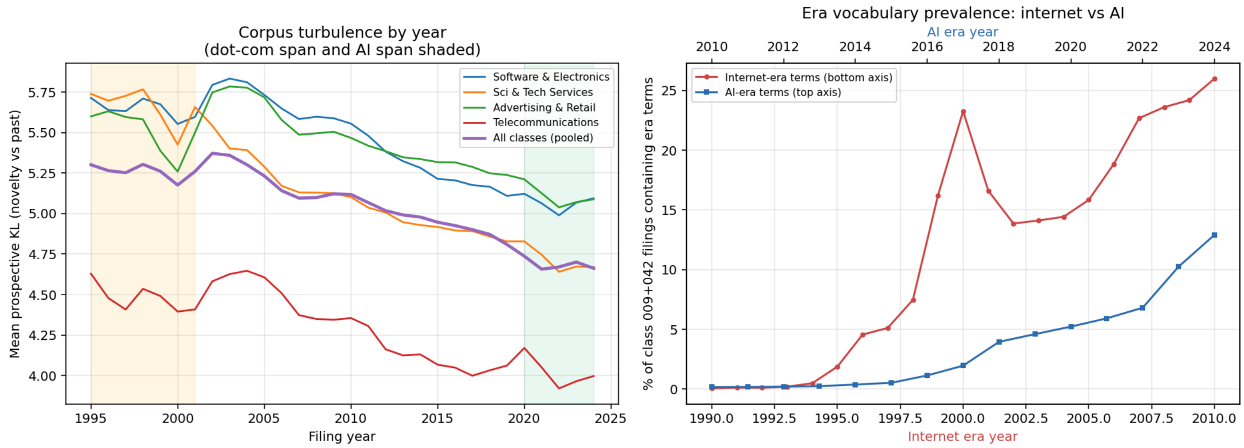

Figure 7: *Left:* per-year mean prospective KL by class and pooled, 1995–2024, with the dot-com and AI spans shaded. *Right:* era-term prevalence in classes 009+042, internet era (bottom axis, from 1994) against AI era (top axis, from 2015).

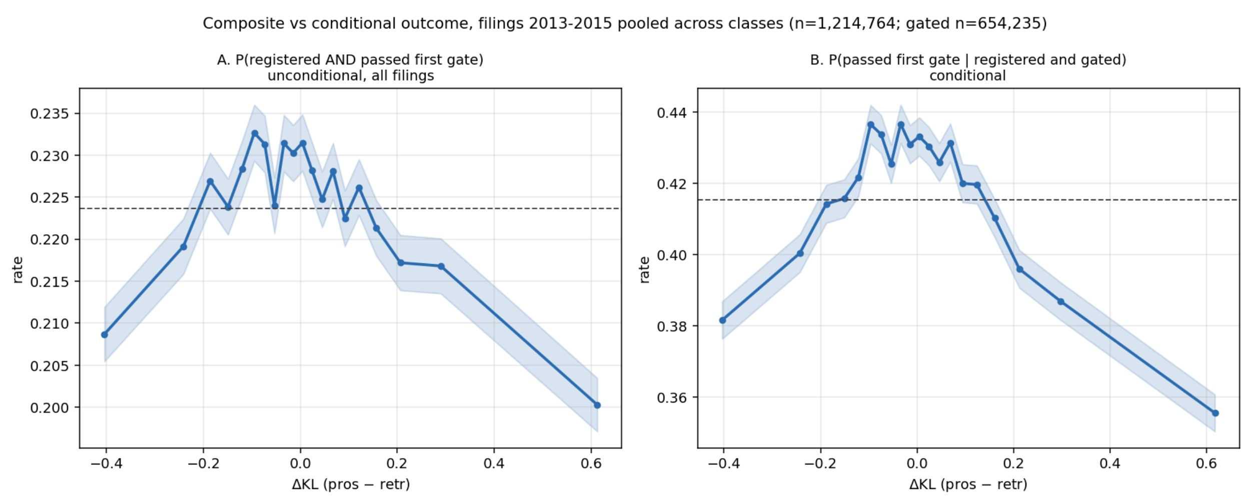

Figure 8: Filings 2013–2015 pooled across classes (n = 1,214,764; gated subset n = 654,235). *A:* the unconditional composite P(registered and passed first gate) over all filings. *B:* P(passed | registered and gated) on the same filings.